

Data Science Research Capstone

Daily Demand Forecasting Using ARIMA Time-Series Modeling

Research Question

How effectively can an ARIMA model forecast future daily demand based on historical demand patterns?

Project Type: Data Science Research / Forecasting

Dataset: Daily demand observations, June 2026

Model: ARIMA(1,1,1)

Evaluation: RMSE and MAPE

Prepared as a research-oriented capstone based on the completed time-series forecasting analysis.

Abstract

Demand forecasting is a practical application of data science in which historical observations are used to estimate future demand. This project studies a 30-day daily demand series covering 1 June 2026 through 30 June 2026. The workflow includes data preparation, exploratory visualization, additive time-series decomposition, stationarity testing, ARIMA modeling, forecast generation, and quantitative evaluation.

The Augmented Dickey-Fuller (ADF) test returned a statistic of -0.9039 with a p-value of 0.7867. Because the p-value is greater than 0.05, the original demand series was treated as non-stationary. An ARIMA(1,1,1) model was therefore fitted using the first 24 observations as training data, while the final 6 observations were retained for testing.

On the six-observation test period, the model produced an RMSE of 14.62 and a MAPE of 6.39%. The fitted model was also used to generate a seven-day future forecast. These results demonstrate a complete and reproducible statistical forecasting workflow while also highlighting the limitations of drawing broad conclusions from a short 30-day series.

Key Findings

- 30 daily observations with no missing values in the three dataset variables.
- The original demand series was non-stationary according to the ADF test.
- ARIMA(1,1,1) was fitted to 24 training observations.
- Test-set RMSE was 14.62 and MAPE was 6.39%.
- A seven-day future forecast was generated from the fitted model.

1. Introduction and Problem Statement

Organizations frequently need estimates of future demand for planning and operational decisions. Historical demand data can contain systematic movement over time as well as recurring variation and irregular noise. Time-series analysis provides a structured way to examine these components and build forecasts without treating observations as independent data points.

The present project focuses on daily demand forecasting. The dataset includes a demand measure together with two contextual indicators: `marketing_event` and `holiday`. The forecasting model in this study uses the historical demand series itself. The contextual indicators are documented as part of the dataset but are not entered as explanatory variables in the ARIMA model used here.

Problem Statement

The central problem is to estimate future daily demand from a short historical sequence. The challenge is to identify whether the series is stationary, represent its temporal structure appropriately, and evaluate forecast performance on observations that were not used to fit the model.

Research Question

How effectively can an ARIMA model forecast future daily demand based on historical demand patterns?

Objectives

- Analyze and visualize historical daily demand.
- Separate observed demand into trend, seasonal, and residual components.
- Test stationarity using the Augmented Dickey-Fuller procedure.
- Fit an ARIMA(1,1,1) forecasting model.
- Evaluate predictions using RMSE and MAPE.
- Produce a seven-day future forecast and discuss practical use and limitations.

2. Dataset Description and Preparation

The dataset contains 30 daily observations from 1 June 2026 through 30 June 2026. It has three variables after the date column is converted to the time-series index: demand, marketing_event, and holiday. All three variables are stored as integer values in the notebook output, and the missing-value check returned zero missing values for each variable.

Variable	Description	Use in study
date	Calendar date of observation	Time-series index
demand	Daily demand value	Forecast target
marketing_event	Indicator for a marketing event	Contextual variable; not modeled
holiday	Indicator for a holiday	Contextual variable; not modeled

The date column was parsed as datetime, observations were sorted chronologically, and date was used as the index. The demand series was then passed to the decomposition, ADF, train/test, and ARIMA procedures.

Descriptive Summary

Statistic	Demand
Count	30
Mean	142.10
Std. deviation	13.84
Minimum	118
25th percentile	132.50
Median	142.00
75th percentile	150.25
Maximum	174

3. Exploratory Data Analysis

The daily demand series was plotted in chronological order before modeling. The observed values range from 118 to 174 during the 30-day period. The sequence shows repeated rises and falls within the month, with higher values appearing around several marketing-event dates. Because the model itself uses demand only, these contextual relationships are descriptive rather than causal conclusions.

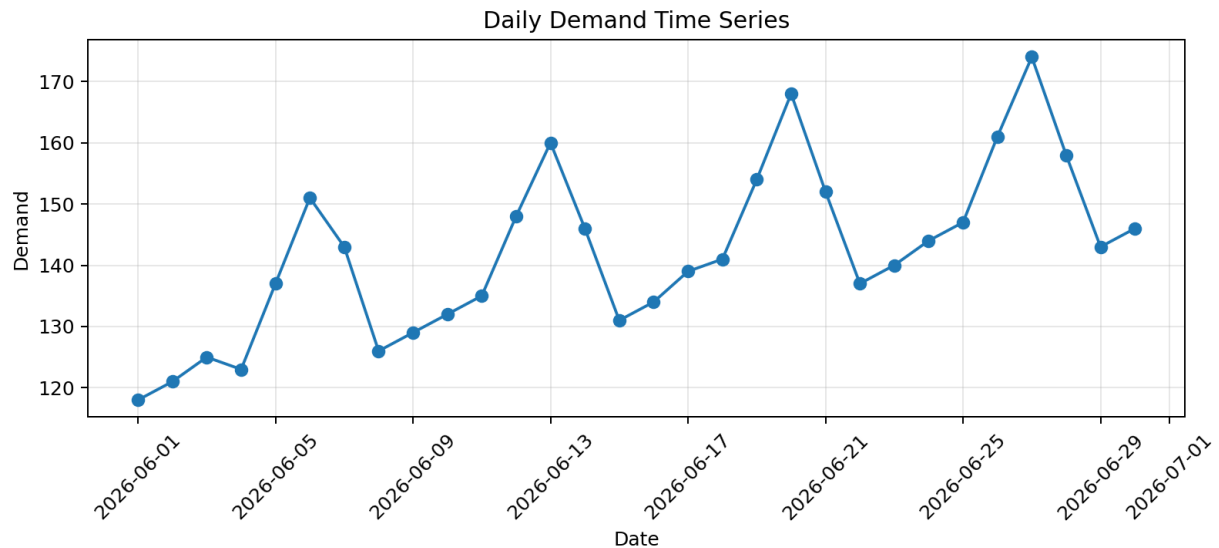


Figure 1. Historical daily demand from 1 June 2026 to 30 June 2026.

Train/Test Design

An 80/20 chronological split was used. The first 24 observations, from 1 June through 24 June, formed the training set. The final 6 observations, from 25 June through 30 June, formed the test set. This preserves temporal ordering and avoids using later observations to fit the model evaluated on earlier data.

4. Time-Series Decomposition

An additive decomposition with a period of seven days was applied to the demand series. The purpose was to inspect the observed series as a combination of trend, seasonal structure, and residual variation. The seven-day period provides a weekly cycle for the decomposition.

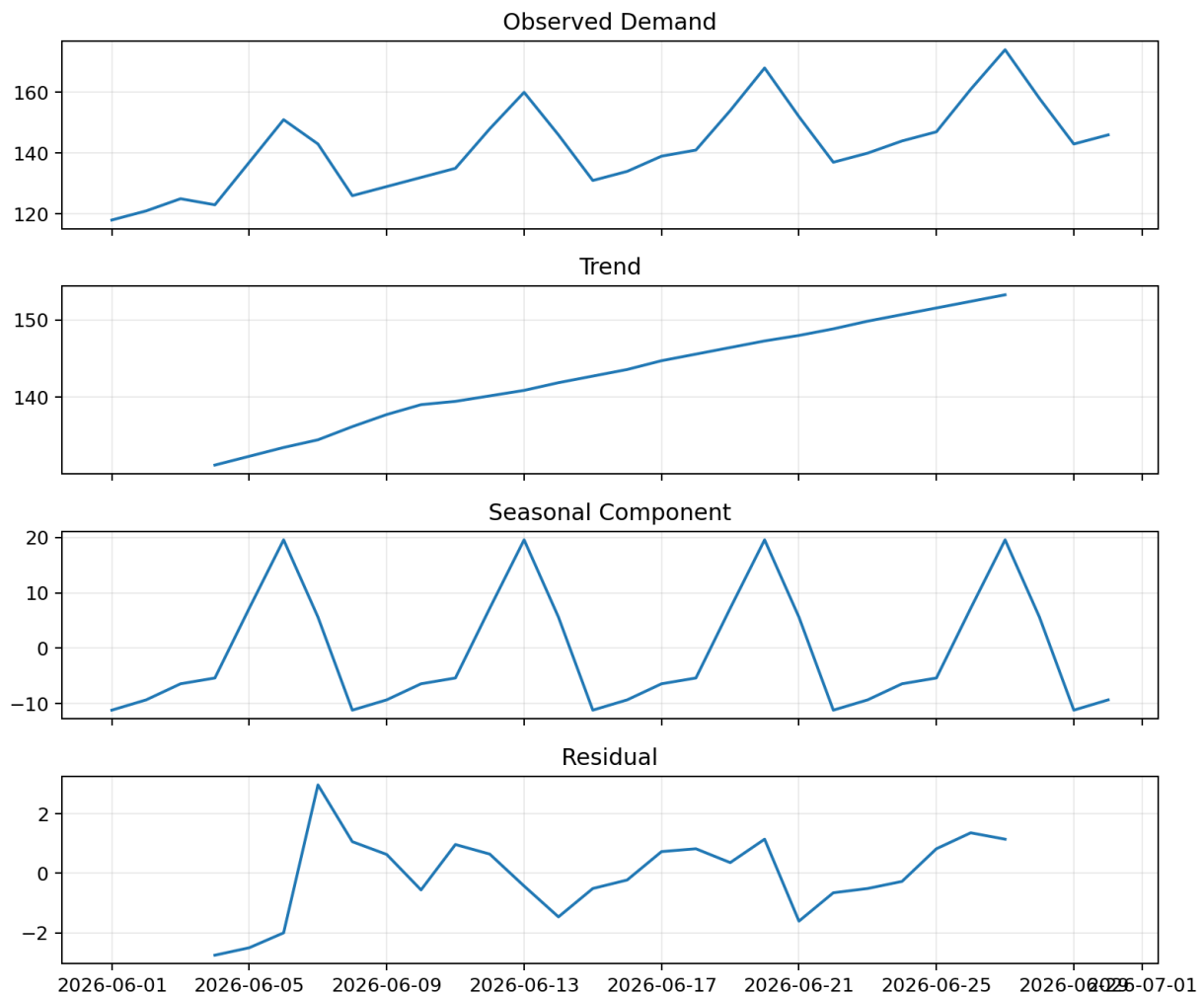


Figure 2. Additive decomposition of observed demand into trend, seasonal, and residual components.

The decomposition is primarily an exploratory diagnostic in this project. It helps describe the temporal structure but does not imply that every component is independently forecast by the ARIMA model.

5. Stationarity Analysis and ARIMA Method

Augmented Dickey-Fuller Test

The Augmented Dickey-Fuller test was applied to the complete demand series. The test statistic was -0.9039 and the p-value was 0.7867. Using a 0.05 significance level, the p-value exceeds the threshold, so the original series was treated as non-stationary.

ARIMA Specification

The forecasting model was specified as ARIMA(1,1,1). The first parameter represents the autoregressive order, the second represents the differencing order, and the third represents the moving-average order. With $d=1$, the model incorporates first differencing to address the non-stationary level of the original series.

Model Fitting

The model was fitted using the 24 training observations. The training-period output reported a log likelihood of approximately -82.216, AIC of 170.433, and BIC of 173.839. The AR coefficient estimate was approximately -0.1460 and the MA coefficient estimate was approximately 0.8065.

Forecast Procedure

After fitting, the model generated six forecasts corresponding to the held-out test period. The same fitted model was subsequently used to produce a seven-day forecast beyond the final observed date.

6. Forecast Results and Evaluation

The forecast was compared with the six observed test-set values. Two standard error measures were calculated: Root Mean Squared Error (RMSE) and Mean Absolute Percentage Error (MAPE).

Metric	Value	Interpretation
RMSE	14.62	Error magnitude in demand units
MAPE	6.39%	Average absolute percentage error

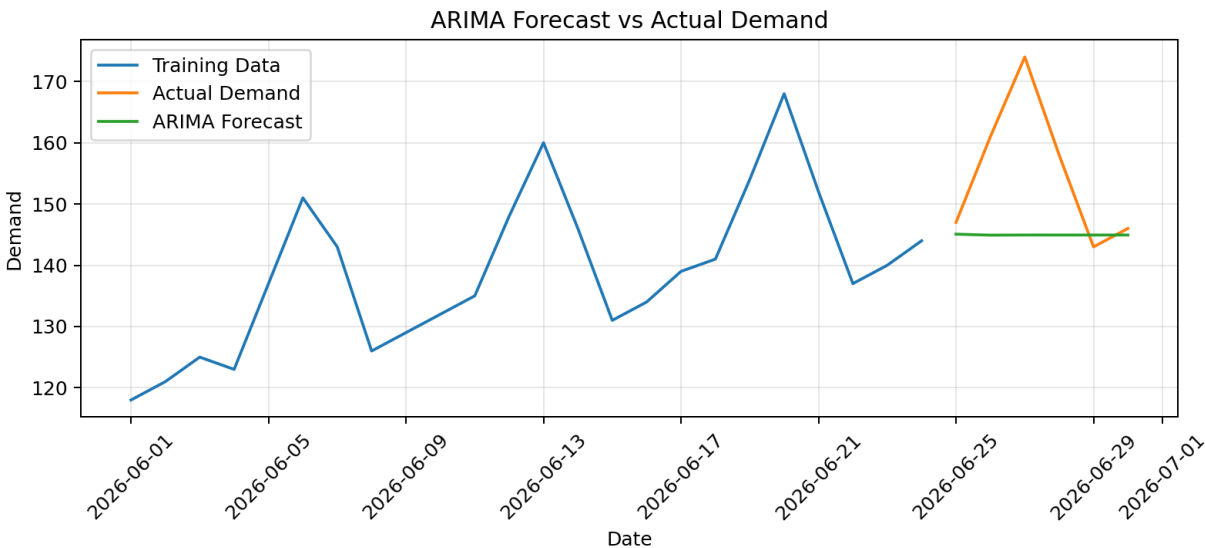


Figure 3. Comparison of training data, actual test demand, and ARIMA forecasts.

RMSE of 14.62 indicates the typical scale of forecast error when expressed in the original demand units. MAPE of 6.39% summarizes the average absolute percentage difference between the forecasts and actual test observations. These metrics describe performance on this particular six-day test period and should not be interpreted as universal performance for other datasets or future periods.

7. Seven-Day Future Forecast

After evaluation, the fitted ARIMA model was used to generate forecasts for the seven days immediately following 30 June 2026. These values are model-based estimates rather than observed outcomes.

Date	Forecasted Demand
2026-07-01	145.07
2026-07-02	144.92
2026-07-03	144.94
2026-07-04	144.93
2026-07-05	144.94
2026-07-06	144.94
2026-07-07	144.94

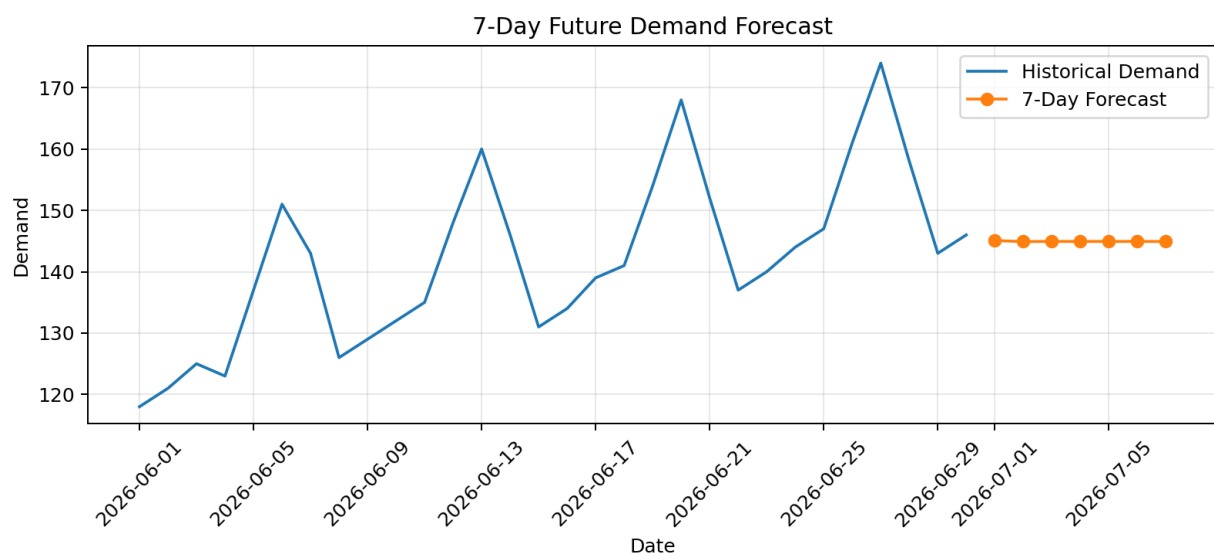


Figure 4. Historical demand followed by the seven-day model forecast.

The forecast can be used as a short-term planning reference. In an operational setting, forecast values should be updated as new observations become available and should be considered alongside relevant external information such as promotions, holidays, supply constraints, or other demand drivers.

8. Practical Implications, Limitations and Future Work

Practical Implications

A short-term demand forecast can support inventory planning, resource allocation, workforce scheduling, and other operational planning activities. The project demonstrates how a reproducible statistical workflow can turn a chronological demand series into measurable forecasts and an explicit error assessment.

Limitations

- The dataset contains only 30 observations, so the study is small and conclusions should not be generalized beyond the available series.
- The ARIMA model uses historical demand only; `marketing_event` and `holiday` are not included as exogenous predictors.
- The test set contains only six observations, so RMSE and MAPE may be sensitive to individual forecast errors.
- Only one ARIMA specification, $ARIMA(1,1,1)$, was evaluated in the notebook; systematic order selection and model comparison could provide a stronger model-selection study.

Future Work

- Expand the historical dataset to cover multiple months or years.
- Compare ARIMA with SARIMA and other forecasting approaches.
- Incorporate marketing events and holidays as external predictors where appropriate.
- Use rolling or walk-forward validation rather than a single six-day test window.
- Compare multiple candidate models using consistent evaluation criteria.

Conclusion

This capstone demonstrated an end-to-end time-series forecasting workflow for daily demand. The series was found to be non-stationary by the ADF test, an $ARIMA(1,1,1)$ model was fitted to the training data, and its forecasts were evaluated on a chronological test set. The resulting RMSE was 14.62 and MAPE was 6.39%. A seven-day future forecast was also generated. The analysis provides a reproducible foundation for demand forecasting while clearly identifying the limitations created by the short dataset and limited model comparison.

9. Reproducibility and References

Software and Analytical Tools

The analysis was implemented in Python using Pandas for data handling, NumPy for numerical operations, Matplotlib and Seaborn for visualization, Statsmodels for decomposition, stationarity testing and ARIMA modeling, and Scikit-learn for forecast-error metrics.

Reproducibility

The project can be reproduced from the supplied daily-demand-series.csv dataset and the associated Jupyter/Colab notebook. The notebook records the preprocessing steps, decomposition, ADF test, train/test split, ARIMA model, forecasts, evaluation metrics, and future forecast.

References

- Box, G. E. P., Jenkins, G. M., Reinsel, G. C., & Ljung, G. M. Time Series Analysis: Forecasting and Control.
- Hyndman, R. J., & Athanasopoulos, G. Forecasting: Principles and Practice.
- Pandas documentation. Data analysis and time-series functionality.
- Statsmodels documentation. Time-series analysis and ARIMA implementation.
- Scikit-learn documentation. Regression and forecasting evaluation metrics.

Project Summary

Dataset: 30 daily observations (1–30 June 2026)

Training observations: 24

Testing observations: 6

Model: ARIMA(1,1,1)

ADF p-value: 0.7867

RMSE: 14.62

MAPE: 6.39%

Future forecast horizon: 7 days